

STUDENT ID	NP069811
INTAKE ID	NP3F2509IT
STUDENT NAME	Ashok Bhattarai

A. Project Title

"Design and Development of an AI-Powered Cloud-Based Public Notice Management System for Nepal"

B. Brief description of project background. (i.e. problem context, rationale, description of problem area, nature of challenge)

Problem Context

Public notices and press releases are an essential part of Nepali civic communication. Government tests are scheduled, job vacancies released, tenders announced, and policy updates and official statements are made very frequently. Those notices, however, were published to dozens of different Government Websites, each with its formatting, layout, and publication mechanism. Unfortunately, there doesn't exist any single central place that has all the necessary information a citizen would require at a time.

The situation has put users, such as students, job seekers, and professionals, in an awkward position where one has to keep checking a number of different government portals for information manually. It is very time-consuming and totally ineffective. Sadly, most of them are in an unstructured form, either long-form text, scanned images, or PDFs, making it quite hard to search through quickly or to understand easily. Most times, it is a student trying to find an exam schedule or a job seeker waiting for vacancy notices that will find themselves going through very long documents for just a single piece of information.

Currently in Nepal, there is no intelligent system to effectively manage public announcements. This absence often makes it difficult for official information to reach the appropriate audience due to inaccessibility problems. This is a clear gap in the public information infrastructure of Nepal which the project seeks to fill.

Rationale

The increased implementation of AI and cloud computing presents a new phase in which a unifying system can be created to solve the problem of fragmentation. The end information will be neat, structured, and searchable, with only a single gateway for users to access information.

The project will develop, design a system, for a Proof of Concept (PoC). The PoC will be applied on five to six important government websites of Nepal so that it is shown that it works in reality. There is also an optional feature of Retrieval Augmented Generation (RAG), where a user can upload his notice-related files in PDF or image format and be able to ask questions about it in plain language. This will give users, straight answers, without needing to read the whole document.

Tangible Benefits

- Citizens should be able to find public notices for multiple sources in the government in one place, saving time and avoiding frustrations.
- Summaries generated by AI have reduced the reading load to help users make sense of a notice without going through an entire document.
- Searchable and filterable database will enable to find the specific notice types quickly.
- RAG allows users to ask questions directly about uploaded documents and get instant answers.

Intangible Benefits

- The gap between the government and the public, specifically for rural and less-connected users, was considerably reduced due to this.
- Increased civic awareness in official information due to availability and clearance of access.
- Increased trust in digital services provided by the government by making available a reliable and intelligent lodging platform.
- Body resulting in empowerment of students and job seekers dependent on timely access to official announcements.

Nature of Challenge

Many were the serious obstacles encountered along the way in developing this system. For one, government websites in Nepal are very inconsistent in terms of their structure, making the development of reliable scrapers a daunting technical task. In addition, a large number of notices get published as scanned PDFs or images that first require OCR to be done for further processing. Reasonable system design is required for a coherent and responsive system that integrates AI components in classification, summarization, and RAG. Fourthly, yet another level of complexity is added to ensure ethical scraping by respecting rate limits and not overloading the servers. Last but not least, ongoing monitoring and adjustment of the system for the updating of government websites is needed to maintain accuracy and relevance.

C. Brief description of project objectives. (i.e. scope of proposal and deliverables)

Deliverables

The system will offer a web platform based on AI, cloud-based and allowing users to search, browse and engage with the public notices gathered on the Nepal government websites. The main system functions that are going to be fulfilled prior to the delivery are structured as follows:

Admin

- An admin can manage and monitor scraping tasks per government website source.
- An admin can review, approve, or discard wrongly scraped or duplicated notices.
- Admin can set up notice categories and AI classification labels.
- Admin can manage user access and system settings.
- An admin is able to view system health, and processing reports.

General Users (Students, Job Seekers, Professionals)

- Enable users to be able to search and filter public notices by category, date, keyword, and source website.
- Users should be allowed to see AI-generated summaries of the notices without having to go through the full document.
- User can set the alert as per their preference and requirements.
- Users should be able to upload their PDF or image files, ask questions using natural language.
- Obtain a clear and accurate answer from the document that they upload.

System (Automated)

- The system will automatically scrap the 6 selected public websites related to notices and press releases.
- The text can be extracted from image or PDF using the OCR.
- The system will automatically categorize the information based on the type of information like notices, press release, job vacancy, tender or else.
- It will generate the brief and clear information out of the long, detailed notice provided by the government websites.

- The system will be designed in such a way that enables accurate retrieval of required information.

D. Brief outline of the resources necessity for this project (i.e. software, user involvement, hardware, access to information/expertise, etc.)

Hardware

The minimum hardware requirement for successful development of this system and meet the project objective are listed below: -

- Processor: Intel Core i5 (or equivalent), 2.4 GHz
- Random Access Memory (RAM): 8 GB
- Storage: 256 GB SSD
- Keyboard, Mouse, and Monitor
- Stable Internet Connection (for cloud access, scraping, and API calls)

Software

The development and execution of the project require the minimum software requirements as follows. All of the tools and platforms used are free and open-source or free tier:

Backend and API Processing

- NestJS: A structured, back-end framework that is scalable in the development of an API for RESTful services. Also, it includes routing, module, and service logic.
- Python 3.x: It uses distinct microservice scripts to handle AI and scraping tasks in conjunction with the NestJS backdrop. It is open-source and free.

- Scrapy and BeautifulSoup4:
A web scraper that automatically retrieves notifications from government websites.
It is open-source and free.
- Selenium with ChromeDriver: This deals with the scraping of JavaScript-rendered pages where that is necessary.
- Tesseract OCR For OCR- scanned PDFs and images of notices. It is free and open-source.
- Hugging Face Transformers: Used to run open-source NLP models with a focus on summarizing and classifying text. Once more, it is open-source and free.
- Ollama: Running open-source, large language models like LLaMA 3 and Mistral locally for the RAG inference feature. It is free and open-source.
- LangChain: Constructing the free and open-source RAG pipeline, which connects documents, embeddings, and the language model.
- ChromaDB: To store and retrieve the vector embeddings used in the RAG document interaction module, free and open-source once again.

Frontend

- Next.js: It is a UI framework for React js for frontend design.
- Tailwind CSS: Probably the most productive way to style a user interface.
- TypeScript: It is a typed superset of JavaScript, something that can work on both the frontend and the backend. This gives rise to more reliable code.

Database and Cloud Deployment

- Amazon RDS and Amazon S3
- Hosting on AWS : Deploy servers and database on AWS.
- Git & GitHub: Version control and source code management. It comes for free.

Documentation and Planning

- Microsoft Word, Excel, Draw.io

Access to Information/Expertise

Consultation and guidance will be accomplished through domain supervisors specializing in artificial intelligence, cloud computing, and software engineering. It proposes to review and apply the technical documentation of AWS and the library and framework documentation of NestJS, Next.js, Hugging Face, LangChain, Ollama, and Scrapy. It proposes to research papers related to NLP, web scraping, and public information systems. More research will, of course, be done on government websites in Nepal for the main data source.

User Involvement

The system would be designed in a manner most suitable for its target users, mainly students, job seekers, and average citizens who are reliant on the very prompt access of government notices. I shall collect feedback from these stakeholder groups during the process of usability testing of the interface on how well their needs are catered for effectively. This input will be used for the fine-tuning of search functionality, notice presentation, and AI summary quality before the final delivery.

E. Academic research being carried out and other information, and techniques being learned. (i.e. what are the names of books you are going to read / data sets you are going to use)

To carry out the deliverables, the preliminary list of books and online resources I will study are as follows:

Books

- **Name:** Natural Language Processing with Transformers
Author: Lewis Tunstall, Leandro von Werra, Thomas Wolf
Publisher: O'Reilly Media
- **Name:** Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow (3rd Edition)
Author: Aurélien Géron
Publisher: O'Reilly Media
- **Name:** Cloud Computing: Concepts, Technology and Architecture
Author: Thomas Erl, Zaigham Mahmood, Ricardo Puttini
Publisher: Prentice Hall
- **Name:** Flask Web Development: Developing Web Applications with Python (2nd Edition)
Author: Miguel Grinberg
Publisher: O'Reilly Media

Online Resources

- Minaee, Shervin; Kalchbrenner, Nal; Cambria, Erik; Nikzad, Niki; Hashemi, Mehdi S., "Deep Learning Based Text Classification: A Comprehensive Review", ACM Computing Surveys, Volume 54
- Lewis, Patrick; Perez, Ethan; Piktus, Aleksandra; Petroni, Fabio; Karpukhin, Vladimir; Goyal, Naman; Küttler, Heinrich; Lewis, Mike; Xiong, Wenjing; Kozareva, Zornitsa; Riedel, Sebastian., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks", Advances in Neural Information Processing Systems (NeurIPS 2020)

- Wirtz, Weyerer and Geyer, "Artificial Intelligence and the Public Sector: Applications and Challenges", International Journal of Public Administration, Volume 42
- Janssen, Charalabidis & Zuiderwijk, "Benefits, Adoption Barriers and Myths of Open Data and Open Government", Information Systems Management, Volume 29
- Mell and Grance, "The NIST Definition of Cloud Computing", NIST Special Publication 800-145

F. Brief description of the development plan for the proposed project. (i.e. which software methodology and why, the major areas of functions to be developed, and the order in which developed)

System Development Methodology

A system development methodology refers to an organized process of planning, running, and controlling the development process. In this project, Agile will be adopted because it is incremental and iterative and thus the most appropriate framework to make use of. System itself combines web scraping and AI processing with cloud deployment and user interface, which need to be continuously improved. Agile allows the development, testing, and refining of each component in the narrow cycles. This will minimize risks, eliminate large scale problems in the late stages, and help maintain a steady feedback that can be used to attain the intended goals.

The phases of development can be broken down into:

- Phase 1: Planning and Architecture: The final needs of the system will be defined in this phase. The general architecture of the system will be designed, and the development environment set up. As such, this includes the GitHub repository, the Supabase database, the NestJS backend scaffold, and the Next.js frontend scaffold.

- Phase 2: Web Scraping Module: Create and test web scrapers with the help of Scrapy and BeautifulSoup for all the 5-6 selected government websites of Nepal.
- Phase 3: Data Store and Processing Pipeline: Design and implement the schema for the database hosted on the cloud infrastructure and a pipeline responsible for storing and organizing the scraped notice data.
- Phase 4: AI Classification and Summarization: Integrate the NLP models into the system to classify the notices in an automated manner and give their summary in a very concise form.
- Phase 5: Web Interface: Now work on Next.js frontend dashboard UI/UX so that users can search, filter, and make themselves conversant with the notices fetched on the screen; linked to NestJS backend API.
- Phase 6: RAG Document Interaction Module: Incorporate a feature to upload documents and ask questions in natural language, programmed through LangChain and the language model API.
- Phase 7: Testing and Deployment: One can expect every possible type of testing to take place during this testing and deployment phase. All identified issues will be fixed, and the final system will be deployed that is, the Next.js frontend on Vercel, the NestJS backend on AWS and the database on RDS.
- Phase 8: Documentation: Prepare the full project report, system documentation, and user guides.

G. Brief description of the evaluation and test plan for the proposed project. (i.e. what are the success criteria and how will be evaluated and implementation will be tested, indicate the estimated size of the demonstration/test database)

Success Criteria

In order to be successful, the AI-empowered, Cloud-Based Public Notice Management System shall have to scrape and securely store public notices from a minimum of five government websites of Nepal. The AI will have to correctly classify notices and produce clear, concise summaries automatically. The system will provide an easy-to-use search interface, optionally with an RAG feature for document-based question answering. It should go through rigorous testing to ensure reliability before getting delivered.

Unit Testing

Unit testing is done to ensure that every single component of the system functions properly. Seldom chosen government website documents are used to test scrapers to ensure they extract the right data. The AI classification and summarization modules are tested to have the right outputs on sample notices. Postman is used to test API endpoints to ensure that they correctly handle requests and offer them in the correct formats. The system should properly categorize the notices and vacancy and examination should not be confused.

Integration Testing

Individual components are tested then combined to make the system work as a whole. Integration testing ensures that the processing pipeline is operating properly and properly moving scraped data into the cloud database and can be properly retrieved by the web interface. As an illustration, the AI module must automatically summarize and categorize a new notice that is gathered by a scraper. The users should be provided with an opportunity to search, read, and view the notice without making mistakes, missing information, and problems with formatting.

Usability Testing

Real students, job seekers, and citizens will be subjected to the system under normal conditioning. The web interface will allow the participants to navigate the interface by

searching a vacancy notice, reading the AI generated summary, and uploading a document to pose a question. Structured questionnaires will also be used to gather the feedback with the ease of use, clarity of information, accuracy of searches, and general satisfaction. The results will be used in making ultimate amendments before delivery of the project.

Test Database

The demonstration and test databases will be populated with around 200 to 300 of the basic public notices which are available on the websites of the Nepal government. Out of these, there will be an adequate distribution across all defined categories: Vacancy, Examination, Tender, Policy Update, General Announcement. This was to test classification, summarization, and search across the full spectrum of content types the system would process in production.